Arpan Roy Chowdhury

Forward Deployed Engineer

+91 9804933234 | arcxx1995@gmail.com | arpanroychowdhury.me | github.com/arcxx1995 | Bengaluru, India

Forward deployed engineer with 6+ years of software engineering experience. Works directly with customers to run technical discovery and deploy production AI solutions, such as RAG pipelines and agentic workflows. Builds with the Claude Agent SDK, MCP, LangChain, and LlamaIndex, and has built voice agents with ElevenLabs. Works in Python and TypeScript, with REST integrations, webhooks, OAuth, and PostgreSQL. Deploys containerized services on AWS.

Experience

Shindigs Kolkata

Forward Deployed Engineer *Nov 2026 – Present*

- Designed and built an event-ticketing platform end to end, from backend APIs and a Postgres data model on Supabase to a Next.js frontend, and run it in production.
- Onboarded customers and built the organizer console around their pain points, with custom features for events that had different needs.
- Built the booking, pricing, and payment flows.
- Built an LLM discovery agent on OpenRouter that recommends events from the live catalogue, with structured outputs, sanitized organizer text to resist prompt injection, a bounded index that stops invented IDs, and a fallback model.

viAct

Software Engineer *Dec 2024 – Aug 2026*

- Worked at the intersection of engineering and customer-facing teams for an enterprise AI workplace-safety platform (computer vision, edge AI, AIoT), building proofs-of-concept and demos to validate technical approaches for prospective customers.
- Built real-time Python data pipelines that ingest streams from edge devices and computer-vision models, with PostGIS for geospatial data.
- Built REST APIs (Flask, FastAPI) and webhooks so customers could connect the platform to their own systems.
- Worked with computer-vision models using PyTorch, ONNX, and TensorRT on the edge AI platform.
- Provisioned AWS infrastructure (EC2, RDS, CloudFront) with Terraform and deployed containerized services in Docker across cloud environments.
- Traced production faults across application code, data pipelines, and infrastructure to the root cause.

Hilti Asia Pacific

SDE *Jul 2022 – Dec 2024*

- Built and maintained backend services and APIs managing customer entitlements, subscriptions, and commercial state for an enterprise platform.
- Contributed to systems automating order intake, validation, and fulfillment across multiple channels and product lines.
- Took part in open design discussions and code reviews, and supported platform reliability through on-call rotation for critical issues.

ETML (ET Media Labs) India

Executive Engineer *Mar 2020 – Jul 2022*

- Built backend systems, automation, and reporting tooling for multi-channel data acquisition and analytics, integrating platform APIs (Google Analytics, Looker Studio).

Projects

ninjatest.app | *Next.js, Supabase, PostgreSQL, Vercel AI SDK* Under development

- Solo-building a real-time 1v1 exam-prep platform with about 45 Postgres security-definer RPCs using row-level locking, and an agentic coaching chat with multi-step tool calling, model fallback, and per-route token and cost limits.

glassbottles.app | *Next.js, Supabase, Meta Cloud API* Open source

- Launched within a month and reached 100+ users and 10+ daily active users, with voice snippets and WhatsApp notifications. Built with an agentic build system of six agent personas run through Claude Code.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL
Agents & AI: Claude Agent SDK, MCP, Vercel AI SDK, LangChain, LangGraph, LlamaIndex, RAG, Voice Agents (ElevenLabs), Claude Code
Backend & Frontend: Node.js, Express.js, Next.js, React, Django REST Framework, Flask, FastAPI, REST APIs, Webhooks, OAuth
Data & Infra: PostgreSQL, pgvector, PostGIS, Supabase, AWS, Terraform, Docker, Linux, CI/CD
Practices: Testing (pytest), Debugging, Code Review, Technical Documentation

Education

IIT Kharagpur

Executive Post Graduate Certificate in Generative AI & Agentic AI *2025*

University of Calcutta

Bachelor of Computer Application (BCA) *Jul 2016 – Aug 2019*